LMA - Assignment 1 Report
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Link to github repo with code: https://github.com/BhavyaV05/LMA-A1.git

1. Architecture Comparison

A bi-encoder encodes the query and each chunk separately into fixed vectors. To check the relevance, we do cosine similarity of the two vectors. Chunk vectors are computed once and stored, so each query needs one forward pass plus one vector search to find them. But, the drawback is that query and chunk never "see" each other.And, all the information about a chunk has to be squeezed into one single vector representaiton, and details such as negation, an exception clause or a specific number can get averaged away.

A cross-encoder takes the query and one chunk as a single input ([CLS] query [SEP] chunk). Self-attention runs across all tokens of both, so each query word can attend directly to each chunk word. The output is a single relevance score for that pair. This is more accurate on fine-grained matching, but nothing can be precomputed. Every (query, chunk) pair needs its own forward pass at query time.

2. Qualitative and Rank Shift analysis

#	Category	Query	Gold chunk (page)	Baseline rank	Re-ranked rank	In top 3 (base → rerank)
1	Exceptions & corner cases	Does Meta retain account data when a user deletes their account?	30 (p14)	7	**1**	No → Yes
2	Buried chunk	What happens to a user's profile and photos when they delete their Facebook account?	30 (p14)	10	**1**	No → Yes
3	Buried chunk, specific detail	What is the deadline to send feedback on the paper and to which address?	43 (p20)	7	**1**	No → Yes
4	Negation & constraints	Which account signal helps spot a hacked account but is not useful for deciding whether a photo is nudity?	6 (p5)	7	**1**	No → Yes
5	Keyword trap / distractor	How much hate speech did Meta remove in late 2021?	48 (p23)	6	**1**	No → Yes
6	Control	Which privacy principle says data should be kept only as long as required, and what is the retention period for hate speech?	22 (p11)	2	4	Yes → No

Following are the top 3 chunk ids.

#	Baseline	Re-ranked
1	31, 29, 14	30, 31, 14
2	32, 56, 37	30, 37, 32
3	49, 10, 58	43, 3, 51
4	25, 26, 55	6, 26, 31
5	23, 24, 54	48, 54, 53
6	21, 22, 17	15, 21, 17

Analysis:
Each of these queries was chosen to represent a particular problem. The categories are mentioned in the first table of this section. All of these were picked as these issues can be fixed with re- ranking and we can examine the difference here. Here is a more detailed

explanation of each query:

#	Gold content the answer should contain	Answer 1 (baseline)	Answer 2 (re-ranked)
1	Normally deleted, but kept for child-safety-disabled accounts	**Refusal**: "The provided context does not answer this."	**Correct.** Gives both the exception (data from child-safety-disabled accounts is retained to stop known violators returning) and the rule (not typically retained for deleted, non-violating accounts). The opening "Yes" is loose phrasing, since the normal case is deletion.
2	Profile, photos and everything added are permanently deleted	**Refusal**	**Correct**: profile, photos and everything added to the account are permanently deleted.
3	`privacy4integrity@fb.com`, through September 30, 2022	**Refusal**	**Correct**: September 30, 2022, `privacy4integrity@fb.com`.
4	A sign-in from a new location or device	**Wrong**: "the user's age", which appears in none of the retrieved chunks	**Correct**: "an account sign-in from a new location or device".
5	17.4 million pieces, October to December 2021	**Refusal**	**Correct**: 17.4 million pieces of hate speech content from October to December 2021.
6	Data Retention principle, two years for hate speech	**Partial.** The period is right (two years), but the principle is wrong ("data minimization").	**Partial.** The principle is right (Data Retention, from chunk 15), but it says the period is not specified, which is true only of its own top 3.

We can see here that:

- The rank shift turns directly into answer quality. For queries 1–5 the gold chunk was missing from the baseline's top 3, and the baseline produced no correct answer on any of them. But the re-ranked pipeline answered all five correctly with the gold chunk at rank 1.
- Refusal is the common baseline failure. On queries 1, 2, 3 and 5 the model followed the prompt and said the context does not answer the question. That is the safe failure. On query 4 it invented an answer ("the user's age").Its top 3 were all about nudity detection, so it had plausible-looking but irrelevant context and filled the gap. This is the more dangerous failure, because a user cannot tell it apart from a real answer.
- Query 1 needs the whole chunk, not just its main sentence. The exception sits at the start of chunk 30, phrased indirectly ("justified in using account information from child safety-disabled accounts"). Once the chunk was retrieved, the model used both the exception and the rule.
- Query 6 (control) shows the trade-off in a two-part question. Each top-3 set holds evidence for one half only. The baseline has the "two years" fact (chunk 22), but chunk 21 cuts off just after the "Data Retention" heading, right after a passage on minimising personal data, so the model named the wrong principle. The re-ranked set has the principle list (chunk 15) but has lost chunk 22. Neither is complete; the re-ranked answer is the more honest of the two, because it states what is missing instead of mislabelling it.
- Overall: fully correct answers rose from 0 of 6 (baseline) to 5 of 6 (re-ranked). The baseline produced 4 refusals, 1 fabrication and 1 partial; the re-ranked pipeline's only shortfall is the control query. With 6 queries and one model, this is illustrative, not statistically meaningful.

3. Latency Profiling

Query #	Baseline (ms)	Re-rank step (ms)	Pipeline 2 total (ms)	Slowdown
1	6.9	683.1	690.0	100.4×
2	5.1	293.9	299.1	58.1×
3	6.8	680.9	687.8	100.6×
4	7.4	594.3	601.7	81.5×
5	7.4	522.4	529.8	72.1×
6	6.6	541.0	547.6	83.0×
Mean	**6.7**	**552.6**	**559.3**	**83.5×**

- Re-ranking 10 chunks costs about 0.55 s on the CPU, roughly 84× the baseline retrieval. The spread between queries (294–683 ms) shows the numbers are noisy.
- The baseline is fast because the corpus is tiny (59 vectors) and the chunk embeddings are already computed. Only the query is encoded.
- LLM generation: one call to the local qwen3:8b (query 1, baseline prompt, model already loaded on the GPU) took 636 ms. Prompt length is about the same in both pipelines, so this cost applies equally to each.
- With a GPU-served LLM, re-ranking is no longer small next to generation. End to end, Pipeline 1 is about 0.64 s per query (7 ms retrieval + 636 ms LLM) and Pipeline 2 is about 1.20 s (559 ms retrieval and re-ranking + 636 ms LLM), so re-ranking nearly doubles the response time. Running the cross-encoder on the GPU as well would shrink most of that gap.
- The LLM figure is a single warm call and depends on answer length. A refusal is short, and a longer answer such as Answer 2 for query 1 takes longer.

4. Scaling

A cross-encoder can’t pre-compute anything. Its score is a function of query and chunk together, so each query requires one forward pass per chunk. On a set of N chunks, N transformer passes per query, linear in the size of the collection. From the measurements above, that is about 55 ms per pair on the CPU, so 100,000 chunks would take about an hour and a half per query on this machine.

A bi-encoder embeds each chunk once, off-line. It only encodes the query at query time, and does a nearest neighbor lookup, which approximate indexes do close to sub-linear. This is why the two-stage design is there: the cheap bi-encoder reduces the entire collection to a few candidates (high recall), and the expensive cross-encoder reorders only those few (high precision).

The downside of this design is that the cross-encoder can only improve the first stage retrieval. If the gold chunk is not in the top 10, re-ranking cannot retrieve it.